

Evolution & Natural Selection

Why do living things fit their environments so well? Why does a desert fox have huge ears and an arctic fox have tiny ones? The answer is a slow, powerful process that has shaped life for billions of years. Let's see how species change over time.

 Georgia Standard S7L5

WHAT YOU'LL LEARN

01 What Is Evolution?

02 Genetic Variation

03 Natural Selection

04 Survival of the Fittest

05 Variation and the Environment

06 The Math of Selection

07 The Fossil Record

08 Putting It All Together

5.1 What Is Evolution?

Evolution is the change in the inherited traits of a species over many generations. It doesn't happen to one individual in its lifetime — it happens to a whole population, slowly, over a very long time.

Think about dogs. Every breed today — from tiny Chihuahuas to giant Great Danes — descended from wolves. Over thousands of years, the traits in those populations shifted. That shift is evolution. In nature, the main process driving it is called **natural selection**.

! Did You Know?

Whales evolved from four-legged land mammals that walked back into the sea about 50 million years ago. Their flippers still contain the bones of those ancient "fingers."

X Common Mix-Up

Individuals don't evolve — **populations** do. A single rabbit can't become better camouflaged during its life, but over generations a rabbit population can.

5.2 Genetic Variation

Within any species, individuals are slightly different from one another. These differences are called **variation**, and they're the raw material evolution works with.

Variation comes mostly from the mixing of genes during sexual reproduction, and sometimes from small changes called **mutations**. Some beetles are darker, some lighter. Some giraffes have slightly longer necks. These differences may seem tiny — but they can decide who survives.

🔑 Why Variation Is Essential

If every member of a species were identical, a single change in the environment could wipe them all out. Variation means **some** individuals are likely to have a trait that helps them survive a new challenge.

5.3 Natural Selection

Natural selection is the process where organisms with helpful traits survive and reproduce more than those without them. Over generations, the helpful traits become more common. Charles Darwin first described this idea in 1859.

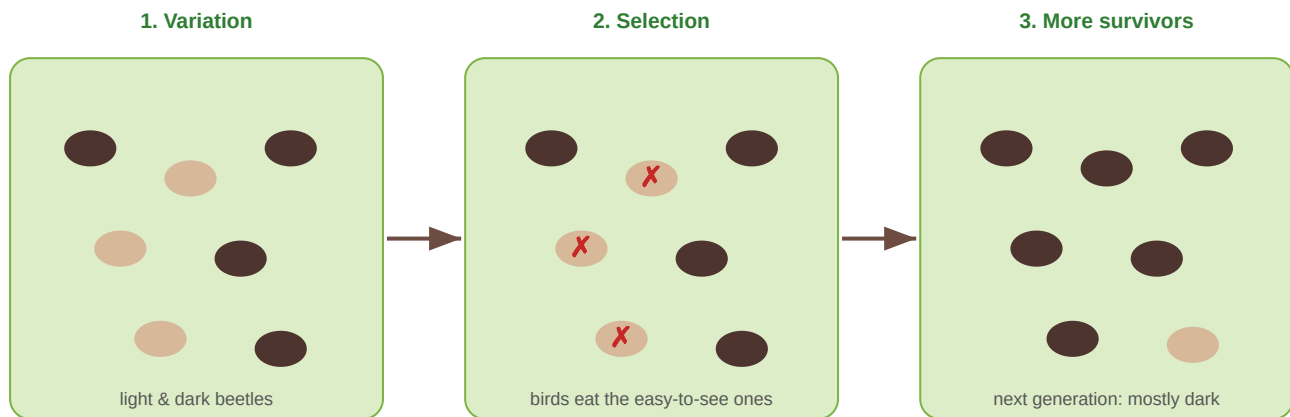


Figure 5.1 — Natural selection in beetles. Dark beetles blend in, survive, and become more common.

Here's the beetle story step by step. A population has both light and dark beetles (**variation**). They live on dark soil, where birds easily spot the light ones and eat them (**selection**). The dark beetles survive and reproduce, passing on the "dark" trait. After many generations, most of the beetles are dark.

★ The Four Steps of Natural Selection

- **Variation:** Individuals differ in their traits.
- **Competition:** More are born than can survive on limited resources.
- **Selection:** Those with helpful traits survive and reproduce.
- **Inheritance:** Survivors pass the helpful traits to offspring.

5.4 Survival of the Fittest

"Fitness" in biology doesn't mean being the strongest or fastest. It means how well an organism is suited to **survive and reproduce** in its environment.

Trait	Why It Increases Fitness
Camouflage	Hides an animal from predators, so it lives to reproduce.
Sharp beak shape	Lets a bird eat the available food more easily.
Resistance to a disease	Survivors of an outbreak pass on resistance.
Bright mating colors	Attracts mates, leading to more offspring.

! A Real Example: Peppered Moths

In 1800s England, most peppered moths were light, matching clean tree bark. When factories darkened the trees with soot, dark moths suddenly had the advantage — birds couldn't see them. Within decades, dark moths became the majority. When pollution dropped, light moths returned. Natural selection in real time.

5.5 Variation and the Environment

The environment decides which traits are helpful. A trait that helps in one place might be useless — or harmful — in another. Selection always depends on the surroundings.

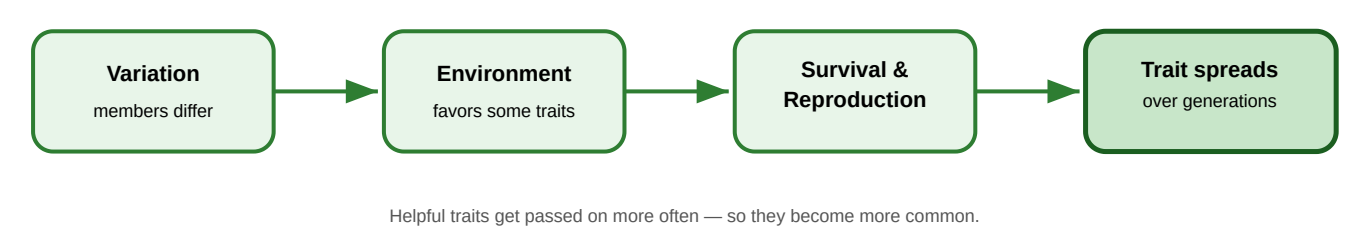


Figure 5.2 — How a helpful trait spreads through a population over time.

Environment	Trait That's Favored
Snowy arctic	White fur for camouflage
Hot desert	Large ears to release heat
Deep dark cave	No eyesight (saves energy), strong other senses
Windy island	Smaller wings, so insects aren't blown out to sea

🔑 Same Trait, Different Outcome

A white coat is a lifesaver for an arctic hare but a death sentence for a hare in a green forest — predators would spot it instantly. The environment is what makes a trait "good" or "bad."

5.6 The Math of Selection

Natural selection can be shown with numbers. When a trait helps survival, the percentage of the population with that trait **increases** each generation. We can track it.

Imagine an island of 100 mice on dark rock. Dark mice hide better, so more survive to reproduce. Watch the percentage of dark mice climb over four generations:

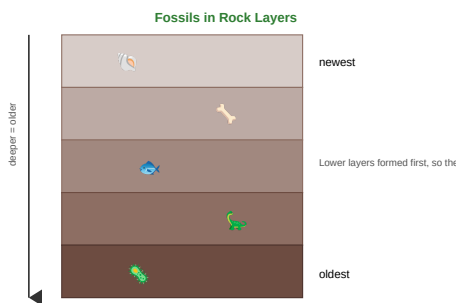
Generation	Dark Mice	Light Mice	% Dark
1	30	70	30%
2	48	52	48%
3	66	34	66%
4	81	19	81%

★ Reading the Pattern

The dark-mouse trait isn't just winning — it's spreading faster and faster because more dark parents means more dark babies. Graphs of natural selection often show this kind of steady climb until the helpful trait dominates the population.

5.7 The Fossil Record

A **fossil** is the preserved remains or trace of a living thing from long ago. Together, all the fossils ever found make up the **fossil record** — our timeline of life on Earth.



Fossils form in layers of rock. Deeper layers formed first, so the fossils in them are **older**. By comparing layers, scientists build a timeline showing how life changed.

The fossil record reveals three big patterns:

- **Existence:** proof that many organisms lived in the past.
- **Diversity:** life has become more varied over time.
- **Extinction:** many species died out and no longer exist.

🔑 Fossils Connect to Today

Fossils show clear links between ancient and modern organisms. We can trace the horse from a dog-sized ancestor, or see feathered dinosaurs that connect to today's birds. The fossil record is powerful evidence for evolution.

5.8 Putting It All Together

Evolution is the change in a species over generations. It runs on variation, and natural selection is its engine: helpful traits survive and spread, while the environment decides what counts as "helpful." We can track these changes with math, and the fossil record gives us the long view of how life has transformed.

Big Idea	What to Remember
Evolution	Inherited traits of a species change over many generations.
Variation	Differences among individuals; the raw material for selection.
Natural selection	Helpful traits survive and reproduce more; they grow common.
Environment	Decides which traits are helpful. Same trait can help or hurt.
The math	A helpful trait's percentage climbs each generation.
Fossil record	Shows existence, diversity, extinction, and links to modern life.

★ **Standard Check — S7L5**

You can now **(a)** use mathematical patterns to show how traits change over generations, **(b)** explain how genetic variation and the environment lead to survival and reproduction, and **(c)** describe the patterns in the fossil record. Standard complete — and that's the whole book!